PROPOSED DESIGN — REV C CONCEPT (v2)

0.914 lb MEETS ≤ 1.00 lb

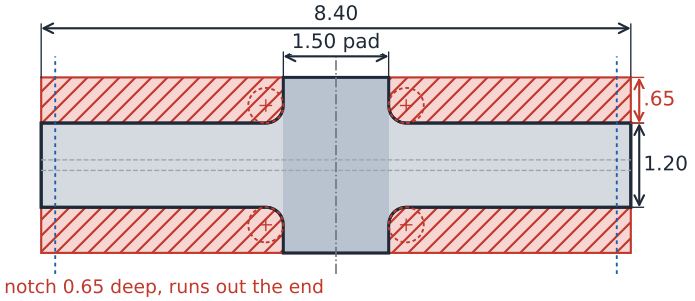
Stock 1 · AA I 3 × 2.03, 6061-T6 · full-width pads at the 3 rollers, flange tips notched elsewhere

EMCH 308 Team 6 · 8.40 in long · 3-point bend, 8.00 in span · inches, from mid-span (CL)

0.086 lb under the limit (Rev B: 0.001 lb)

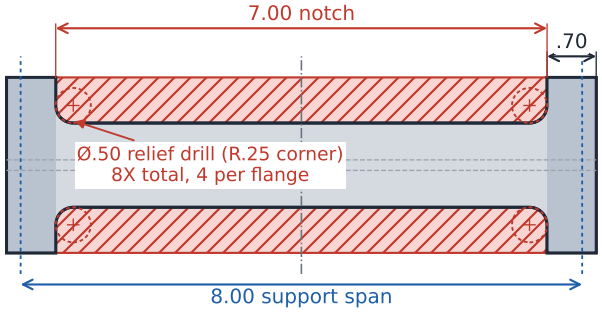
TOP FLANGE

viewed from above · load-roller pad at CL

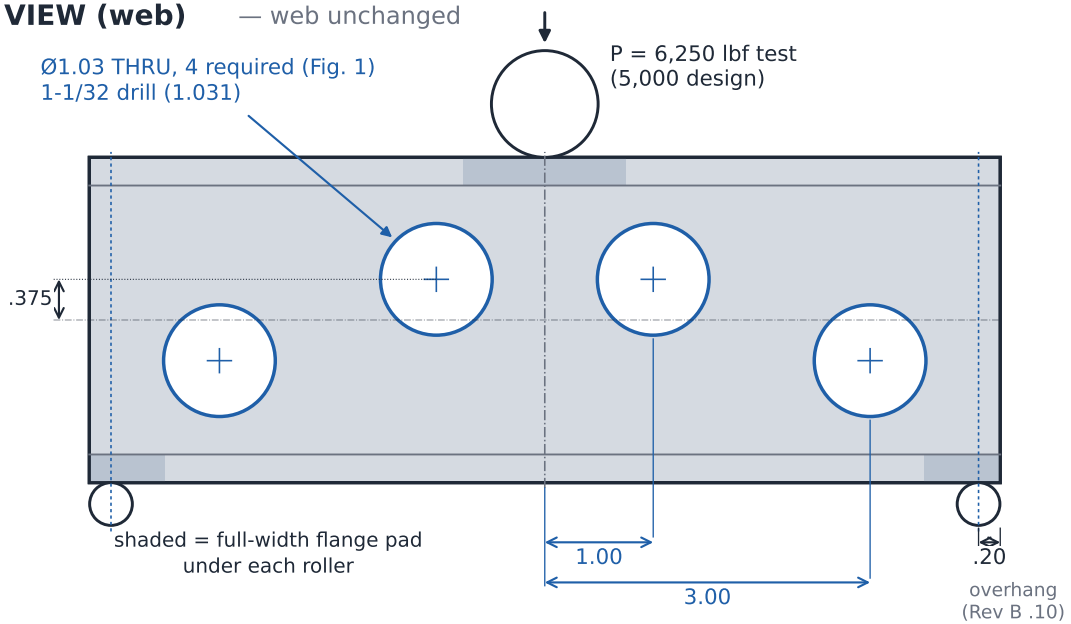


BOTTOM FLANGE

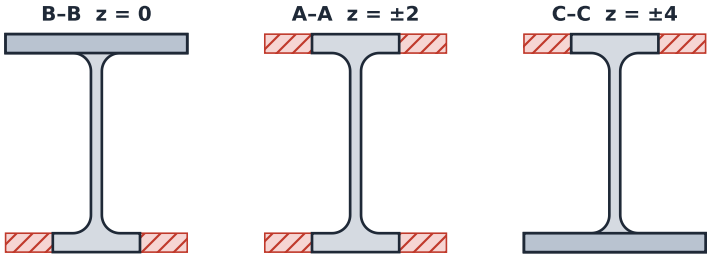
viewed from below · support pads at both ends



SIDE VIEW (web) — web unchanged



SECTIONS — flange .26, web .15, R.25 fillets (stock)





Concept for team review — hand calculations only.
Confirm in Abaqus before a Rev C drawing goes to MTC.

WEIGHT (0.098 lb/in³)

Blank, 8.40 in × 0.1692 lb/in	1.421 lb
– flange notches, 27.8 in × 0.65 × 0.26	–0.458 lb
– 4 required Ø1.03 holes	–0.049 lb
Finished weight	**0.914 lb**
Worst case, every notch 0.02 shallow	0.928 lb

COST (course cost sheet)

Material, Stock 1, 8.40 in raw	$55.02
Saw: set-up + 14 cuts	$40.05
Drill: set-up + 12 holes (4 web, 8 relief)	$36.71
Total (limit $600; Rev B $143.47)	**$131.78**

STRENGTH (hand calc at 6,250 lbf)

Bending at CL, bottom fibre	11.9 ksi → FOS 3.4
Bending at z = 1 hole section	10.5 ksi → FOS 3.8
Avg web shear at Ø1.03 holes	14.4 ksi (VM 24.9)
Hole edge to fillet strip	0.100 in (Rev B .115)

The 0.15 in web sets first yield (about 4,500 lbf in every Stock 1 FE run). Flange tips away from the rollers do not: the notched FE run (0.75 deep, pads kept) matched Rev B's curve. This plan keeps the web whole, keeps every roller zone full width, and doubles the support overhang. FE must confirm.

MANUFACTURING (MTC)

- Drill the 8 Ø.50 relief holes first; each saw cut ends tangent to one, so every inside corner is R.25.
- Top flange: 2 cuts per notch (runs out the end). Bottom flange: 3 cuts per notch. 14 cuts total.
- Two drill sizes: 1-1/32 (web), 1/2 (relief).
- No cross-cut: uses the 8.40 raw length as supplied.
- Break all edges 0.02 max.
- Confirm with David Wilson that MTC can saw to a relief hole this way.